Year 8

REVISION PACK

Unit 3

Name: _____

Class: _____

Unit 3 · Simplifying and expanding

Worked example: collecting like terms and expanding brackets

(a) Simplify $3y^2 + 5y + y^2 + 2$ (b) Expand and simplify $5(2j + 3) - 3(2j + 1)$ (c) Expand and simplify $2(x + 4y) + 5(y - x)$

(a)	**Step 1**	Collect the y^2 terms	$3y^2 + y^2 = 4y^2$
	Step 2	Answer	$4y^2 + 5y + 2$
(b)	**Step 1**	Expand	$10j + 15 - 6j - 3$
	Step 2	Collect like terms	$4j + 12$
(c)	**Step 1**	Expand	$2x + 8y + 5y - 5x$
	Step 2	Collect like terms	$-3x + 13y$

Practice: collecting like terms and expanding brackets

1.1 (a) Simplify $2y^2 + 3y + y^2 + 6$ (b) Expand and simplify $5(3j + 4) - 2(2j + 3)$ (c) Expand and simplify $5(x + 2y) + 3(y - x)$

1.2 (a) Simplify $5a^2 + 2a + 3a^2 + 1$ (b) Expand and simplify $3(4j + 2) - 1(2j + 3)$ (c) Expand and simplify $3(x + 5y) + 2(y - x)$

1.3 (a) Simplify $4m^2 + 7m + 2m^2 + 9$ (b) Expand and simplify $6(2j + 1) - 3(2j + 3)$ (c) Expand and simplify $5(x + 3y) + 4(y - x)$

1.4 (a) Simplify $p^2 + 4p + 6p^2 + 5$ (b) Expand and simplify $2(5j + 4) - 4(2j + 3)$ (c) Expand and simplify $6(x + 2y) + 2(y - x)$

1.5 (a) Simplify $3t^2 + 6t + 4t^2 + 2$ (b) Expand and simplify $5(3j + 2) - 2(2j + 3)$ (c) Expand and simplify $2(x + 7y) + 4(y - x)$

1.6 (a) Simplify $7n^2 + n + 2n^2 + 8$ (b) Expand and simplify $4(2j + 6) - 3(2j + 3)$ (c) Expand and simplify $7(x + 3y) + 5(y - x)$

Unit 3 · Using a formula

Worked example: substituting and solving

$v^2 = u^2 + 2as$. Find s when (a) $u = 10$, $v = 0$, $a = -5$

Step 1	Put the numbers in	$0^2 = 10^2 + 2 \times (-5) \times s$
Step 2	Square, then multiply	$0 = 100 - 10s$
Step 3	Add $10s$	$10s = 100$
Step 4	Answer	$s = 10$

Practice: substituting and solving

2.1 $v^2 = u^2 + 2as$. Find s when (a) $u = 20$, $v = 0$, $a = -4$ (b) $u = 6$, $v = 10$, $a = 2$

2.2 $v^2 = u^2 + 2as$. Find s when (a) $u = 12$, $v = 0$, $a = -3$ (b) $u = 5$, $v = 13$, $a = 4$

2.3 $v^2 = u^2 + 2as$. Find s when (a) $u = 14$, $v = 0$, $a = -7$ (b) $u = 3$, $v = 9$, $a = 6$

2.4 $v^2 = u^2 + 2as$. Find s when (a) $u = 30$, $v = 0$, $a = -10$ (b) $u = 4$, $v = 8$, $a = 3$

2.5 $v^2 = u^2 + 2as$. Find s when (a) $u = 16$, $v = 0$, $a = -8$ (b) $u = 2$, $v = 12$, $a = 5$

2.6 $v^2 = u^2 + 2as$. Find s when (a) $u = 9$, $v = 0$, $a = -2$ (b) $u = 1$, $v = 11$, $a = 10$

Worked example: factorising fully

Factorise fully

(a) $12x + 18$ (b) $3d^2 - 5d$ (c) $9x + xy$ (d) $10x^2 + 4x^3$

(e) Ana writes $8 - 12t = 2(4 - 6t)$. Is this factorised fully? Explain.

(a)	Step 1	HCF of 12 and 18	6
	Step 2	Answer	$6(2x + 3)$
(b)	Step 1	Common factor	d
	Step 2	Answer	$d(3d - 5)$
(c)	Step 1	Common factor	x
	Step 2	Answer	$x(9 + y)$
(d)	Step 1	HCF	$2x^2$
	Step 2	Answer	$2x^2(5 + 2x)$
(e)	Step 1	HCF of 8 and 12 is 4, not 2	$8 - 12t = 4(2 - 3t)$
	Step 2	Answer	No, 4 is the HCF

Practice: factorising fully

3.1 Factorise fully

(a) $20x + 35$ (b) $3d^2 - 7d$ (c) $12x + xy$ (d) $10x^2 + 15x^3$

(e) Ana writes $30 - 20t = 2(15 - 10t)$. Is this factorised fully? Explain.

3.2 Factorise fully

(a) $14y + 21$ (b) $5k^2 - 3k$ (c) $7x + xy$ (d) $6x^2 + 9x^3$

(e) Ana writes $18 + 24a = 3(6 + 8a)$. Is this factorised fully? Explain.

3.3 Factorise fully

(a) $24a + 30$ (b) $4m^2 - 9m$ (c) $11x + xy$ (d) $8x^2 + 20x^3$

(e) Ana writes $20 - 30n = 5(4 - 6n)$. Is this factorised fully? Explain.

3.4 Factorise fully

(a) $10p + 25$ (b) $3h^2 - 8h$ (c) $5x + xy$ (d) $15x^2 + 10x^3$

(e) Ana writes $16x + 24 = 4(4x + 6)$. Is this factorised fully? Explain.

3.5 Factorise fully

(a) $18b + 27$ (b) $6t^2 - 5t$ (c) $20x + xy$ (d) $14x^2 + 21x^3$

(e) Ana writes $36 - 48y = 6(6 - 8y)$. Is this factorised fully? Explain.

3.6 Factorise fully

(a) $16c + 40$ (b) $7w^2 - 2w$ (c) $13x + xy$ (d) $9x^2 + 6x^3$

(e) Ana writes $28 + 42b = 2(14 + 21b)$. Is this factorised fully? Explain.

Worked example: simplifying with the laws of indices

Simplify

(a) $m^5 \times m^2$ (b) $c^9 \div c^4$ (c) $3x^3 \times 5xy$ (d) $(2ab^3)^4$

(a)	Step 1	Add the powers	$5 + 2 = 7$
	Step 2	Answer	m^7
(b)	Step 1	Subtract the powers	$9 - 4 = 5$
	Step 2	Answer	c^5
(c)	Step 1	Multiply the numbers	$3 \times 5 = 15$
	Step 2	Add the powers of x	$x^3 \times x = x^4$
	Step 3	Answer	$15x^4y$
(d)	Step 1	Number to the power 4	$2^4 = 16$
	Step 2	Multiply the powers by 4	$a^4,\ \ b^{12}$
	Step 3	Answer	$16a^4b^{12}$

Practice: simplifying with the laws of indices

4.1 Simplify

(a) $n^6 \times n^3$ (b) $c^{12} \div c^5$ (c) $4x^3 \times 2xy$ (d) $(3ab^2)^3$

4.2 Simplify

(a) $p^2 \times p^7$ (b) $k^{10} \div k^4$ (c) $5x^2 \times 3xy$ (d) $(4ab^3)^2$

4.3 Simplify

(a) $t^4 \times t^4$ (b) $y^8 \div y^3$ (c) $6x^4 \times 2xy$ (d) $(2ab^5)^3$

4.4 Simplify

(a) $h^9 \times h^2$ (b) $d^{11} \div d^6$ (c) $2x^5 \times 7xy$ (d) $(5ab^2)^2$

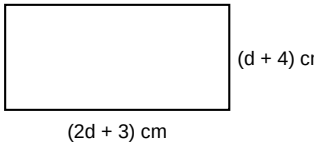
4.5 Simplify

(a) $w^3 \times w^8$ (b) $g^{15} \div g^9$ (c) $3x^2 \times 4xy$ (d) $(3ab^4)^2$

4.6 Simplify

(a) $r^5 \times r^5$ (b) $z^7 \div z^2$ (c) $7x^3 \times 2xy$ (d) $(10ab^2)^3$

Worked example: perimeter and area of a rectangle



The diagram shows a rectangle.

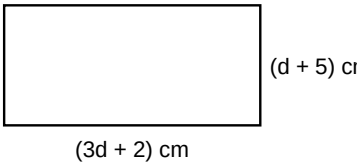
(a) The perimeter is 38 cm. Write and solve an equation to find d.

(b) Find the area of the rectangle.

(a)	Step 1	Add all 4 sides	$2(2d+3)+2(d+4)=38$
	Step 2	Expand and simplify	$6d+14=38$
	Step 3	Subtract 14	$6d=24$
	Step 4	Divide by 6	$d=4$
(b)	Step 1	Length	$2\times4+3=11$
	Step 2	Width	$4+4=8$
	Step 3	Area = length × width	$11\times8=88\text{ cm}^2$

Practice: perimeter and area of a rectangle

5.1

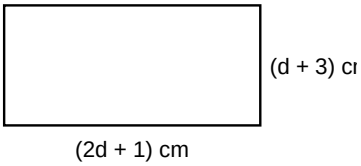


The diagram shows a rectangle.

(a) The perimeter is 30 cm. Write and solve an equation to find d.

(b) Find the area of the rectangle.

5.2

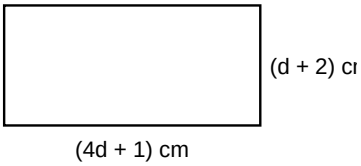


The diagram shows a rectangle.

(a) The perimeter is 32 cm. Write and solve an equation to find d.

(b) Find the area of the rectangle.

5.3



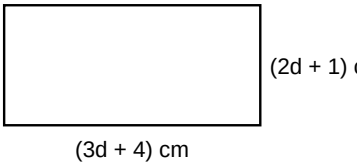
The diagram shows a rectangle.

(a) The perimeter is 36 cm. Write and solve an equation to find d.

(b) Find the area of the rectangle.

Practice: perimeter and area of a rectangle (continued)

5.4

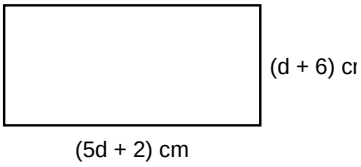


The diagram shows a rectangle.

(a) The perimeter is 50 cm. Write and solve an equation to find d.

(b) Find the area of the rectangle.

5.5

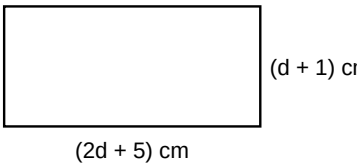


The diagram shows a rectangle.

(a) The perimeter is 52 cm. Write and solve an equation to find d.

(b) Find the area of the rectangle.

5.6



The diagram shows a rectangle.

(a) The perimeter is 30 cm. Write and solve an equation to find d.

(b) Find the area of the rectangle.

Worked example: showing two expressions are the same

Show that

(a) $x^2(x+3x^2)+4x^3 \equiv 5x^3+3x^4$ (b) $3(x+2)+2(x-2) \equiv 5x+2$

(a)	Step 1	Expand	$x^2 \times x + x^2 \times 3x^2 + 4x^3$
	Step 2	Simplify the powers	$x^3 + 3x^4 + 4x^3$
	Step 3	Collect like terms	$5x^3 + 3x^4$
(b)	Step 1	Expand	$3x + 6 + 2x - 4$
	Step 2	Collect like terms	$5x + 2$

Practice: showing two expressions are the same

6.1 Show that

(a) $x^2(x+2x^2)+4x^3 \equiv 5x^3+2x^4$ (b) $4(x+3)+2(x-3) \equiv 6x+6$

6.2 Show that

(a) $x^2(x+4x^2)+2x^3 \equiv 3x^3+4x^4$ (b) $5(x+1)+3(x-1) \equiv 8x+2$

6.3 Show that

(a) $x^2(x+5x^2)+3x^3 \equiv 4x^3+5x^4$ (b) $2(x+4)+3(x-4) \equiv 5x-4$

6.4 Show that

(a) $x^2(x+3x^2)+6x^3 \equiv 7x^3+3x^4$ (b) $6(x+2)+4(x-2) \equiv 10x+4$

6.5 Show that

(a) $x^2(x+6x^2)+2x^3 \equiv 3x^3+6x^4$ (b) $3(x+5)+2(x-5) \equiv 5x+5$

6.6 Show that

(a) $x^2(x+7x^2)+4x^3 \equiv 5x^3+7x^4$ (b) $7(x+1)+5(x-1) \equiv 12x+2$